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Paper 1: Automated Pharmacometric Model Development by Leveraging Low-Dimensional Neural

Bibliographic Information

- **Title**: Automated Pharmacometric Model Development by Leveraging Low-Dimensional Neural ODEs and LASSO Regression
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Paper Type: METHODOLOGY

Quick Take

This paper introduces a fully automated method for pharmacometric model development that uses neural ODEs to learn dynamics and LASSO regression to select explicit functions, eliminating the need for iterative manual model building. The approach is demonstrated on three examples and produces interpretable structural models with comparable or better fit than traditional methods.

Executive Summary

This paper presents a fully automated method for pharmacometric model development that combines low-dimensional neural ordinary differential equations (NODEs) with LASSO regression. The approach first fits a NODE to the data, then uses the trained neural network to generate derivative data, and finally applies LASSO to select a sparse set of explicit functions (e.g., linear, exponential, Emax) that best describe the dynamics. The method is demonstrated on three examples: neonatal weight growth, bi-exponential PK data, and warfarin PK/PD. Results show that the proposed structural models recover meaningful dynamics with comparable or better fit than reference models, while dramatically reducing the need for manual, iterative model building.

Scientific Context & Motivation

Current pharmacometric model development is a manual, iterative process that requires substantial time and resources. Existing automation approaches still rely on iterative fitting of candidate models. Neural ODEs can fit data without specifying a structural model but are black-box. This work addresses the gap by automatically extracting interpretable structural models from NODEs using LASSO regression, providing a bridge between data-driven and mechanistic modeling.

⚡ Methodological Snapshot

The NODE-LASSO approach automates pharmacometric model development by combining neural ordinary differential equations (NODEs) with LASSO regression. In step 1, a low-dimensional NODE (with a neural network on the right-hand side) is fitted to the data using standard mixed-effects modeling software. In step 2, derivative data is generated by evaluating the trained neural network over a range of state values. In step 3, LASSO regression is applied to select a sparse set of explicit functions (e.g., constant, linear, exponential, Emax) that best describe the derivative data. In step 4, the selected functions are assembled into a structural ODE model, which is then fitted to the original data. The key advantage is that model selection (step 3) does not require solving any ODEs, making it computationally efficient.

Statistical Framework

The method assumes that the underlying dynamics can be approximated by a low-dimensional ODE with a neural network right-hand side. The NODE is fitted using maximum likelihood estimation in a nonlinear mixed-effects framework (Monolix). Derivative data is generated by evaluating the trained neural network. LASSO regression with L1 regularization is used for variable selection, assuming that the true derivative can be expressed as a sparse linear combination of candidate functions. The final structural model is fitted using standard PMX methods with log-normal parameter distributions.

Estimator Behavior

The NODE-LASSO approach does not directly estimate parameters in the traditional sense; rather, it identifies functional forms. The final structural model parameters are estimated via standard maximum likelihood in Monolix. The LASSO regression provides consistent variable selection under appropriate regularization (λ chosen via cross-validation). The final model fits showed comparable or better MARE values than the NODE and reference PMX models, indicating no substantial loss in predictive performance.

Validation Design

The method was validated on three simulated/real datasets: (1) neonatal weight growth (simulated from a known model), (2) bi-exponential PK data (simulated from a two-compartment model), (3) warfarin PK/PD (real data). For each, the NODE fit was assessed via goodness-of-fit plots and MARE. The proposed structural model was compared to the NODE and a reference PMX model. No formal simulation-estimation studies or bootstrap were performed, but the method's ability to recover known structures (e.g., two-compartment model) was assessed.

Comparison to Alternatives

Compared to traditional stepwise model building and existing automation algorithms (e.g., SINDy), the NODE-LASSO approach avoids iterative ODE solving during model selection. Unlike SINDy, which requires direct derivative approximations from noisy/sparse data, NODE-LASSO generates clean derivative data from the fitted NODE within a mixed-effects framework. The approach is also more flexible than fixed candidate model libraries used in other automation methods.

Implementation Guidance

Implementation requires the pmxNODE R package (for NODE fitting in Monolix) and the glmnet R package (for LASSO). The candidate function library should include pharmacometrically plausible functions (constant, linear, exponential, Emax). Users should first verify adequate NODE fit (via goodness-of-fit plots) before proceeding. The LASSO regularization parameter λ should be selected via cross-validation. The final structural model can be fitted using standard PMX software (Monolix, NONMEM, nlmixr2). Computational cost is low since ODE solving is only needed for the initial NODE fit and final model fit.

Key Findings

The NODE-LASSO approach successfully proposed structural models for all three examples. For neonatal weight data, the proposed model (linear + exponential terms) fit comparably to the NODE (MARE 0.092 pop, 0.007 ind). For bi-exponential PK data, the proposed model was equivalent to the central compartment of a two-compartment model (MARE 0.269 pop, 0.097 ind vs. reference 0.268/0.088). For warfarin PD, the proposed model (Emax + exponential inhibition) outperformed the traditional Emax model (MARE 0.217 pop, 0.066 ind vs. reference 0.221/0.085). The approach is highly efficient as it avoids iterative ODE solving during model selection.

Strengths & Limitations

Strengths

- Fully automated model development without iterative ODE solving.
- Interpretable structural models derived from black-box NODEs.
- Highly efficient computationally.
- Flexible candidate function library and easy fine-tuning.
- Demonstrated on multiple realistic pharmacometric scenarios.

Limitations (Acknowledged by Authors)

- Low-dimensional NODEs do not model latent states (e.g., peripheral compartments).
- No pharmacological reasoning is directly provided for the proposed structural model.
- The method requires adequate NODE fit before proceeding.

Limitations (Expert Review)

- The candidate function library must be specified by the user, which may bias the selection.
- The method does not automatically handle time-varying parameters or covariates.
- The final model may include terms without clear mechanistic interpretation (e.g., exponential term in warfarin example).

Generalizability

The method is demonstrated on three diverse examples (weight growth, PK, PK/PD) and is expected to generalize to other pharmacometric datasets with similar dynamics. However, performance may degrade for systems with latent states or very sparse/noisy data.

Key Equations

Low-dimensional NODE

$$\frac{dx}{dt} = f(x, t; \theta) = \mathrm{NN}\ (x, t; \phi)$$

Low-dimensional NODE structure with state-dependent neural network for the derivative of the state variable.

Weight model

$$\frac{dw}{dt} = a \cdot w + b \cdot e^{-c \cdot t}$$

Proposed structural model for neonatal weight growth derived from NODE-LASSO, including a linear and an exponential term.

Two-compartment PK model

$$\frac{dC}{dt} = -k_{el} \cdot C - k_{12} \cdot C + k_{21} \cdot C_2$$

Proposed structural model for bi-exponential PK data, which is equivalent to the central compartment of a two-compartment model.

Warfarin PD model

$$\frac{dR}{dt} = k_{in} - k_{out} \cdot R \cdot \left(1 - \frac{E_{\max} \cdot C}{EC_{50} + C} - \alpha \cdot e^{-\beta \cdot t}\right)$$

Proposed structural model for warfarin PD, including an Emax and an exponential inhibition term.

Figures & Tables

- **Figure 1**: Schematic of the four-step NODE-LASSO approach: (1) fit low-dimensional NODE, (2) generate derivative data from NNs, (3) LASSO regression to select functions, (4) fit final structural model.
  - *Significance*: Provides a clear overview of the automated workflow, highlighting that ODE solving is only required in steps 1 and 4.
- **Figure 3**: Population fits for the three example datasets (weight, two-compartment PK, warfarin PD) comparing NODE, proposed structural model, and reference PMX model.
  - *Significance*: Demonstrates that the proposed structural models fit the data comparably to the NODE and reference models, validating the approach.
- **Table 1**: Table of MARE values for population and individual fits across NODE, proposed structural model, and reference PMX model for all three examples.
  - *Significance*: Quantifies the predictive performance, showing that the proposed model achieves similar or better MARE values than the NODE and reference models.

Code & Reproducibility Assessment

The method is implemented using the pmxNODE R package (for NODE fitting in Monolix) and the glmnet R package (for LASSO regression). Code for the examples is not explicitly provided in the paper, but the described workflow is reproducible with these tools.

Future Directions

Future work should extend the NODE-LASSO approach to handle latent states (e.g., peripheral compartments) by using multi-dimensional NODEs. Additionally, incorporating pharmacological knowledge into the candidate function library and developing automated diagnostics for model selection (e.g., based on BIC or cross-validation) would enhance usability. The method could also be applied to more complex scenarios such as disease progression models or systems pharmacology.

Expert Commentary

This work represents a significant step toward automating the most labor-intensive part of PMX modeling—structural model identification. The key innovation is the use of NODEs to generate clean derivative data, which then enables efficient LASSO-based function selection without iterative ODE solving. The method is particularly promising for complex or time-varying dynamics (e.g., maturation, disease progression) where traditional model building is challenging. However, the reliance on low-dimensional NODEs (no latent states) is a limitation for multi-compartment systems, though the two-compartment example shows the method can still recover equivalent dynamics. Future work should extend the approach to handle latent states and provide pharmacological interpretability for the selected functions.

Bottom Line

The NODE-LASSO approach offers a fully automated, computationally efficient alternative to iterative PMX model building. It is best suited for scenarios where the underlying dynamics are unknown or complex, but users must verify NODE fit quality and carefully select candidate functions. The method excels at proposing interpretable structural models from data without requiring prior mechanistic knowledge, though final models may need pharmacological refinement.

Figures

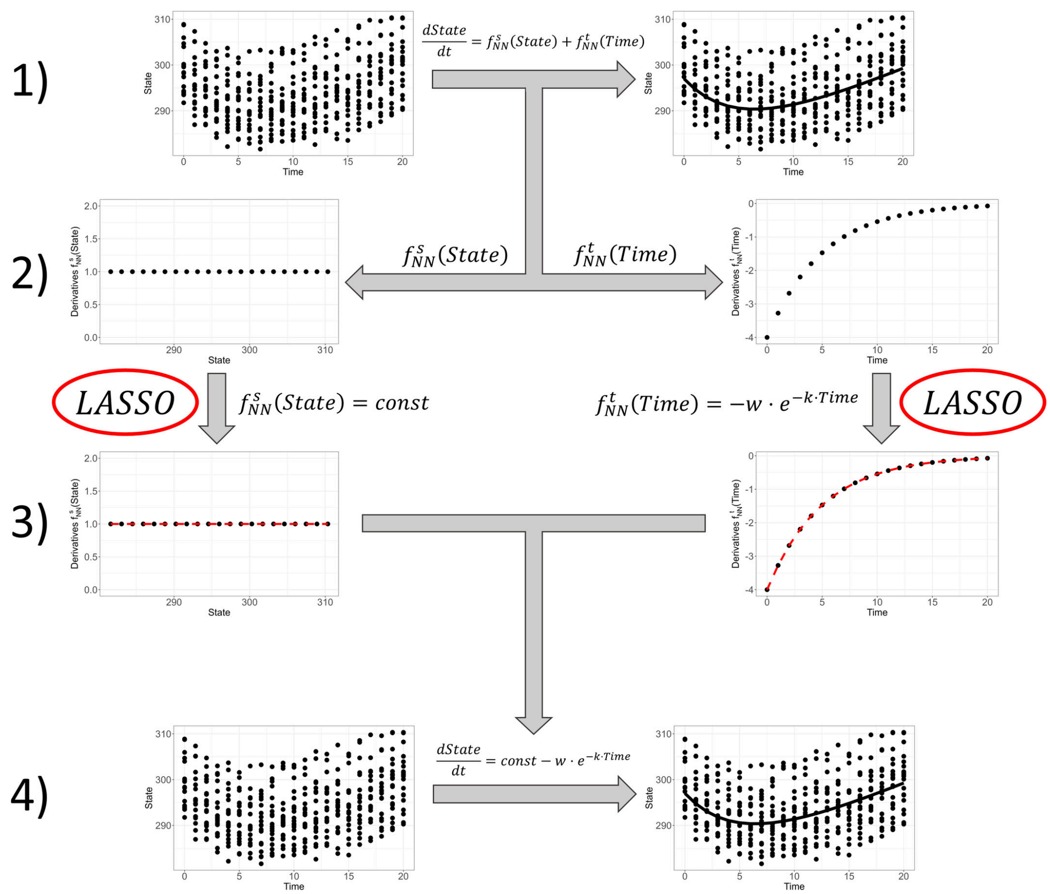

Figure 1: Overview of the general NODE-LASSO approach with (1) fitting a low-dimensional NODE to the data, (2) generating derivative data from the NNs, (3) utilizing LASSO